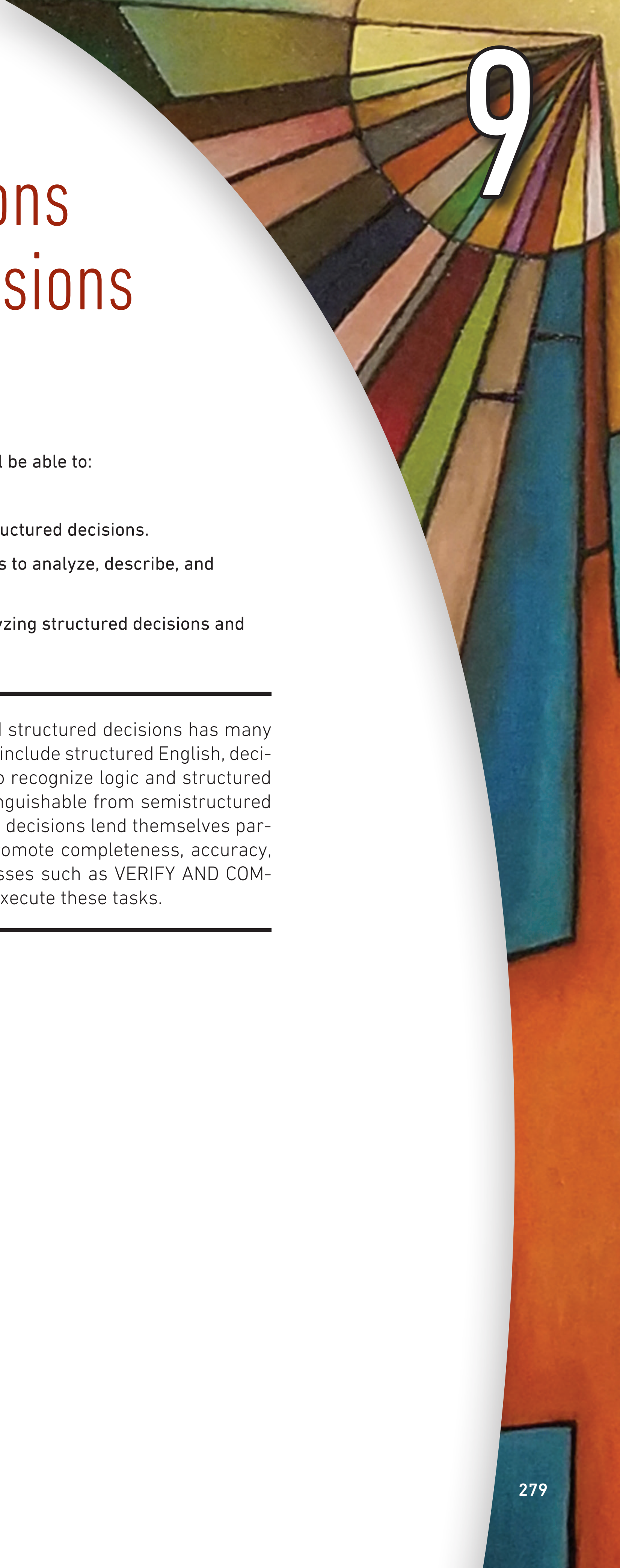

9

Process Specifications and Structured Decisions

LEARNING OBJECTIVES

Once you have mastered the material in this chapter, you will be able to:

1. Understand the purpose of process specifications.
2. Recognize the difference between structured and semistructured decisions.
3. Use structured English, decision tables, and decision trees to analyze, describe, and document structured decisions.
4. Choose an appropriate decision analysis method for analyzing structured decisions and creating process specifications.

A systems analyst approaching process specifications and structured decisions has many options for documenting and analyzing them. The methods include structured English, decision tables, and decision trees. It is important to be able to recognize logic and structured decisions that occur in a business and how they are distinguishable from semistructured decisions that tend to involve human judgment. Structured decisions lend themselves particularly well to analysis with systematic methods that promote completeness, accuracy, and communication. In Chapters 7 and 8 we noted processes such as VERIFY AND COMPUTE FEES, but we did not explain the logic necessary to execute these tasks.

CONSULTING OPPORTUNITY 9.1

Kit Chen Kaboodle, Inc.

"I don't want to get anyone stirred up, but I think we need to sift through our unfilled order policies," says Kit Chen. "I wouldn't want to put a strain on our customers. As you know already, Kit Chen Kaboodle is a Web and mail-order cookware business specializing in 'klassy kitsch for kitchens,' as our latest catalog says. I mean, we've got everything you need to do gourmet cooking and entertaining: nutmeg grinders, potato whisks, egg separators, turkey basters, placemats with cats on 'em, ice cube trays in shamrock shapes, and more."

"Here's how we've been handling unfilled orders. We search our unfilled orders file from the Internet as well as mail-order sales once a week. If the order was filled this week, we delete the record, and the rest is gravy. If we haven't written to the customer in four weeks, we send 'em this cute card with a chef peeking into the oven, saying, 'Not ready yet.' (It's a notification that their item is still on backorder.)"

"If the backorder date changed to greater than 45 days from now, we send out a notice. If the merchandise is seasonal (as with Halloween treat bags, Christmas cookie cutters, or Valentine's Day cake molds) and the backorder date is 30 days or more, though, we send out a notice with a chef glaring at his egg timer."

"If the backorder date changed at all and we haven't sent out a card in the last two weeks, we send out a card with a chef checking his recipe. If the merchandise is no longer available, we send a notice (complete with chef crying in the corner) and delete the record. We haven't begun to use email in place of mailed cards, but I'd like to.

"Thanks for listening to all this. I think we've got the right ingredients for a good policy; we just need to blend them together and cook up something special."

You are the systems analyst Kit hired. Go through the narrative of how Kit Chen Kaboodle, Inc. handles unfilled orders, drawing boxes around each action Kit mentions and circling each condition brought up. Make a list of any ambiguities you would like to clarify in a later interview, and then write five questions to address them.

Overview of Process Specifications

To determine the human information requirements of a decision analysis strategy, a systems analyst must first determine the users' objectives, along with the organization's objectives, using either a top-down approach or an object-oriented approach. The systems analyst must understand the principles of organizations and have a working knowledge of data-gathering techniques. The top-down approach is critical because all human decisions in the organization should be related, at least indirectly, to the broad objectives of the entire organization.

Process specifications—sometimes called *minispecs* because they are a small portion of the total project specifications—are created for primitive processes on a data flow diagram as well as for some higher-level processes that explode to a child diagram. They also may be created for class methods in object-oriented design, and, in a more general sense, for the steps in a use case (as discussed in Chapters 2 and 10). These specifications explain the decision-making logic and formulas that will transform process input data into output. Each derived element must have process logic to show how it is produced from the base elements or other previously created derived elements that are input to the primitive process.

The three goals of producing process specifications are as follows:

1. To reduce the ambiguity of the process. This goal compels the analyst to learn details about how the process works. Any vague areas should be noted, written down, and consolidated for all process specifications. These observations form a basis and provide the questions for follow-up interviews with the user community.
2. To obtain a precise description of what is accomplished. This is usually included in a packet of specifications for the programmer.
3. To validate the system design. This goal includes ensuring that a process has all the input data flow necessary for producing the output. In addition, all input and output must be represented on the data flow diagram.

You will find many situations in which process specifications are not created. Sometimes the process is very simple or the computer code already exists. This eventuality would be noted in the process description, and no further design would be required. Categories of processes that generally *do not* require specifications are as follows:

1. Processes that represent physical input or output, such as read and write. These processes usually require only simple logic.
2. Processes that represent simple data validation, which is usually fairly easy to accomplish. The edit criteria are included in the data dictionary and incorporated in the computer source code. Process specifications may be produced for complex editing.
3. Processes that use prewritten code. These processes are generally included in a system as procedures, methods, and functions or in class libraries (that are either purchased or available free on the Web).

These blocks are computer program code that is stored on the computer system. They usually perform a general system function, such as validating a date or a check digit. These general-purpose subprograms are written and documented only once but form a series of building blocks that may be used in many systems throughout the organization. Thus, these subprograms appear as processes on many data flow diagrams (or as class methods discussed in Chapter 10).

Process Specification Format

Process specifications link a process to a data flow diagram, and hence a data dictionary, as illustrated in Figure 9.1. Each process specification should be entered on a separate form or into a CASE (computer assisted software engineering) tool screen such as the one used by the CASE tool Visible Analyst. Enter the following information:

1. The process number, which must match the process ID on the data flow diagram. This specification allows an analyst to work on or review any process and to locate the data flow diagram containing the process easily.
2. The process name, which must be the same as the name displayed in the process symbol on the data flow diagram.
3. A brief description of what the process accomplishes.
4. A list of input data flows, using the names found on the data flow diagram. Data names used in the formula or logic should match those in the data dictionary to ensure consistency and good communication.
5. The output data flows, also using the data flow diagram and data dictionary names.
6. An indication of the type of process: batch, online, or manual. All online processes require screen designs, and all manual processes should have well-defined procedures for employees performing the process tasks.

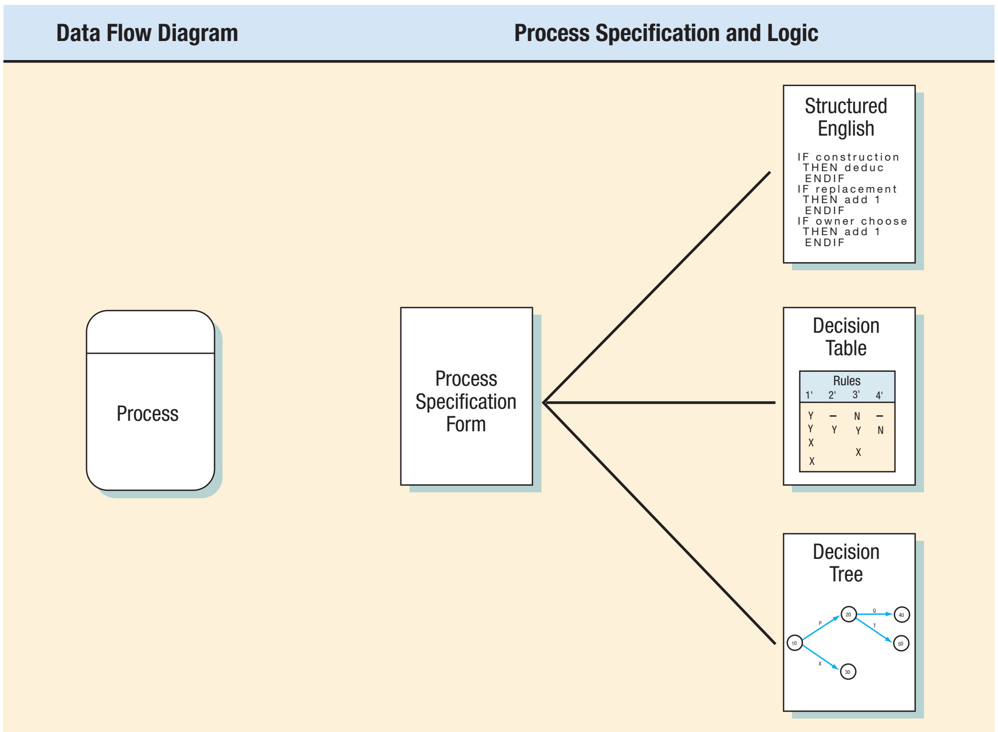

FIGURE 9.1

How process specifications relate to a data flow diagram.

7. If the process uses prewritten code, include the name of the subprogram or function containing that code.
8. A description of the process logic that states policy and business rules in everyday language, not computer language pseudo-code. Business rules are the procedures, or perhaps a set of conditions or formulas, that allow a corporation to run its business. The early problem definition (as explained in Chapter 3) that you completed initially may provide a starting place for this description. Common business rule formats include the following:

   - Definitions of business terms
   - Business conditions and actions
   - Data integrity constraints
   - Mathematical and functional derivations
   - Logical inferences
   - Processing sequences
   - Relationships among facts about the business

9. If there is not enough room on the form for a complete structured English description, or if there is a decision table or tree depicting the logic, include the corresponding table or tree name.
10. A list of any unresolved issues, incomplete portions of logic, or other concerns. These issues form the basis of the questions used for follow-up interviews with users or business experts you have added to your project team.

These items should be entered to complete a process specification form, which includes a process number, process name, or both from the data flow diagram, as well as the eight other items shown in the World's Trend example (Figure 9.2). Notice that completing this form thoroughly facilitates linking the process to the data flow diagram and the data dictionary.

Structured English

When process logic involves formulas or iteration, or when structured decisions are not complex, an appropriate technique for analyzing the decision process is the use of structured English. As the name implies, structured English is based on (1) structured logic, or instructions organized into nested and grouped procedures, and (2) simple English statements such as add, multiply, and move. A word problem can be transformed into structured English by putting the decision rules into their proper sequence and using the convention of IF-THEN-ELSE statements throughout.

Writing Structured English

To write structured English, you may want to use the following conventions:

1. Express all logic in terms of one of these four types: sequential structures, decision structures, case structures, or iterations (see Figure 9.3 for examples).
2. Use and capitalize accepted keywords such as IF, THEN, ELSE, DO, DO WHILE, DO UNTIL, and PERFORM.
3. Indent blocks of statements to show their hierarchy (nesting) clearly.
4. When words or phrases have been defined in a data dictionary (as in Chapter 8), underline those words or phrases to signify that they have a specialized, reserved meaning.
5. Be careful when using "and" and "or," and avoid confusion when distinguishing between "greater than" and "greater than or equal to" and similar relationships. "A and B" means both A and B; "A or B" means either A or B, but not both. Clarify the logical statements now rather than waiting until the program coding stage.

A STRUCTURED ENGLISH EXAMPLE The following example demonstrates how a spoken procedure for processing medical claims is transformed into structured English:

> We process all our claims in this manner. First, we determine whether the claimant has ever sent in a claim before; if not, we set up a new record. The claim totals for

Process Specification Form

Number 1.3
Name Determine Quantity Available
Description Determine if an item is available for sale. If it is not available, create a backordered item record. Determine the quantity available.

Input Data Flow
Valid item from Process 1.2
Quantity on Hand from Item Record

Output Data Flow
Available Item (Item Number + Quantity Sold) to Processes 1.4 & 1.5
Backordered item to Inventory Control

Type of Process
☑ Online ☐ Batch ☐ Manual

Subprogram/Function Name

Process Logic:
IF the Order Item Quantity is greater than Quantity on Hand
    Then Move Order Item Quantity to Available Item Quantity
        Move Order Item Number to Available Item Number
ELSE
    Subtract Quantity on Hand from Order Item Quantity
        giving Quantity Backordered
    Move Quantity Backordered to Backordered Item Record
    Move Item Number to Backordered Item Record
    DO write Backordered Record
    Move Quantity on Hand to Available Item Quantity
    Move Order Item Number to Available Item Number
ENDIF

Refer to: Name: ____
☐ Structured English ☐ Decision Table ☐ Decision Tree

Unresolved Issues: Should the amount that is on order for this item be taken into account? Would this, combined with the expected arrival date of goods on order, change how the quantity available is calculated?

FIGURE 9.2

An example of a completed process specification form for determining whether an item is available.

the year are then updated. Next, we determine if a claimant has policy A or policy B, which differ in deductibles and copayments (the percentage of the claim claimants pay themselves). For both policies, we check to see if the deductible has been met ($100 for policy A and $50 for policy B). If the deductible has not been met, we apply the claim to the deductible. Another step adjusts for the copayment; we subtract the percentage the claimant pays (40 percent for policy A and 60 percent for policy B) from the claim. Then we issue a check if there is money coming to the claimant, print a summary of the transaction, and update our accounts. We do this until all claims for that day are processed.

CONSULTING OPPORTUNITY 9.2

Kneading Structure

Kit Chen has risen to the occasion and answered your questions concerning the policy for handling unfilled orders at Kit Chen Kaboodle, Inc. Based on those answers and any assumptions you need to make, pour Kit's narrative (from Consulting Opportunity 9.1) into a new mold by rewriting the recipe for handling unfilled orders in structured English. In a paragraph, describe how this process might change if you used email for notification rather than regular mail sent by post.

In examining the foregoing statements, we notice some simple sequence structures, particularly at the beginning and end. There are a couple of decision structures, and it is most appropriate to nest them, first by determining which policy (A or B) to use and then by subtracting the correct deductibles and copayments. The last sentence points to an iteration: Either DO UNTIL all the claims are processed or DO WHILE there are claims remaining.

Realizing that it is possible to nest the decision structures according to policy plans, we can write the structured English for the foregoing example (see Figure 9.4). As we begin to work on the structured English, we find that some logic and relationships that seemed clear at one time are actually ambiguous. For example, do we add the claim to the year-to-date (YTD) claim before or after updating the deductible? Is it possible that an error can occur if something other than policy A or B is stored in the claimant's record? We subtract 40 percent of what from the claim? These ambiguities need to be clarified at this point.

Besides the obvious advantage of clarifying the logic and relationships found in human languages, structured English has another important advantage: it is a communication tool.

FIGURE 9.3

Examples of logic expressed in a sequential structure, a decision structure, a case structure, and an iteration.

Structured English Type	Example
Sequential Structure A block of instructions in which no branching occurs	Action #1 Action #2 Action #3
Decision Structure Only IF a condition is true, complete the following statements; otherwise, jump to the ELSE	IF Condition A is True THEN implement Action A ELSE implement Action B ENDIF
Case Structure A special type of decision structure in which the cases are mutually exclusive (if one occurs, the others cannot)	IF Case #1 Implement Action #1 ELSE IF Case #2 Implement Action #2 ELSE IF Case #3 Implement Action #3 ELSE IF Case #4 Implement Action #4 ELSE print error ENDIF
Iteration Blocks of statements that are repeated until done	DO WHILE there are customers. Action #1 ENDDO

```
DO WHILE there are claims remaining
  IF claimant has not sent in a claim
    THEN set up new claimant record
  ELSE continue
Add claim to YTD Claim
  IF claimant has policy–plan A
    THEN IF deductible of $100.00 has not been met
        THEN subtract deductible–not–met from claim
        Update deductible
      ELSE continue
      ENDIF
      Subtract copayment of 40% of claim from claim
  ELSE IF claimant has policy–plan B.
      THEN IF deductible of $50.00 has not been met
          THEN subtract deductible–not–met from claim
          Update deductible
        ELSE continue
        ENDIF
        Subtract copayment of 60% of claim from claim
      ELSE continue
    ELSE write plan–error–message
    ENDIF
  ENDIF
  IF claim is greater than zero
    THEN print check
  ENDIF
  Print summary for claimant
  Update accounts
ENDDO
```

FIGURE 9.4

Structured English for the medical-claim processing system. Underlining signifies that the terms have been defined in the data dictionary.

Structured English can be taught to and hence understood by users in the organization, so if communication is important, structured English is a viable alternative for decision analysis.

Data Dictionary and Process Specifications

All computer programs may be coded using the three basic constructs: sequence, selection (IF . . . THEN . . . ELSE and the case structure), and iteration or looping. The data dictionary indicates which of these constructs must be included in the process specifications.

If the data dictionary for the input and output data flow contains a series of fields without any iteration—{ }—or selection—[ ]—the process specification will contain a simple sequence of statements, such as MOVE, ADD, and SUBTRACT. Refer to the example of a data dictionary for the SHIPPING STATEMENT, illustrated in Figure 9.5. Notice that the data dictionary for the SHIPPING STATEMENT has the ORDER NUMBER, ORDER DATE, and CUSTOMER NUMBER as simple sequential fields. The corresponding logic, shown in lines 3 through 5 in the corresponding structured English in Figure 9.6, consists of simple MOVE statements.

A data structure with optional elements contained in parentheses or either/or elements contained in brackets will have a corresponding IF . . . THEN . . . ELSE statement in the process specification. Also, if an amount, such as QUANTITY BACKORDERED, is greater than zero, the underlying logic will be IF . . . THEN . . . ELSE. Iteration, indicated by braces on a data structure, must have a corresponding DO WHILE, DO UNTIL, or PERFORM UNTIL to control looping on the process specification. The data structure for the

FIGURE 9.5

Data structure for a shipping statement for World's Trend.

Shipping Statement =	Order Number +
	Order Date +
	Customer Number +
	Customer Name +
	Customer Address +
	$^{5}_{1}\{$Order Item Lines$\}$ +
	Number of Items +
	Merchandise Total +
	(Tax) +
	Shipping and Handling +
	Order Total
Customer Name =	First Name +
	(Middle Initial) +
	Last Name
Address =	Street +
	(Apartment) +
	City +
	State +
	Zip +
	(Zip Expansion) +
	(Country)
Order Item Lines =	Item Number +
	Quantity Ordered +
	Quantity Backordered +
	Item Description +
	Size Description +
	Color Description +
	Unit Price +
	Extended Amount

ORDER ITEM LINES allows up to five items in the loop. Lines 8 through 17 show the statements contained in the DO WHILE through the END DO necessary to produce the multiple ORDER ITEM LINES.

Decision Tables

A decision table is a table of rows and columns, separated into four quadrants, as shown in Figure 9.7. The upper-left quadrant contains the condition(s); the upper-right quadrant contains the condition alternatives. The lower half of the table contains the actions to be taken on the left and the rules for executing the actions on the right. When a decision table is used to determine which action needs to be taken, the logic moves clockwise, beginning from the upper left.

Suppose a store wanted to illustrate its policy on noncash customer purchases. The company could do so using a simple decision table, as shown in Figure 9.8. Each of the three conditions (sale under $50, pays by check, and uses credit cards) has only two alternatives. The two alternatives are Y (yes, it is true) or N (no, it is not true). Four actions are possible:

1. Complete the sale after verifying the signature.
2. Complete the sale. No signature needed.
3. Call the supervisor for approval.
4. Communicate electronically with the bank for credit card authorization.

Structured English

Format the Shipping Statement. After each line of the statement has been formatted, write the shipping line.

```
1.  GET Order Record
2.  GET Customer Record
3.  Move Order Number to shipping statement
4.  Move Order Date to Shipping Statement
5.  Move Customer Number to Shipping Statement
6.  DO format Customer Name (leave only one space between First/Middle/Last)
7.  DO format Customer Address lines
8.  DO WHILE there are items for the order
9.      GET Item Record
10.     DO Format Item Line
11.     Multiply Unit Price by Quantity Ordered giving Extended Amount
12.     Move Extended Amount to Order Item Lines
13.     Add Extended Amount to Merchandise Total
14.     IF Quantity Backordered is greater than zero
15.         Move Quantity Backordered to Order Item Lines
16.     ENDIF
17. ENDDO
18. Move Merchandise Total to Shipping Statement
19. Move 0 to Tax
20. IF State is equal to CT
21.     Multiply Merchandise Total by Tax Rate giving Tax
22. ENDIF
23. Move Tax to Shipping Statement
24. DO calculate Shipping and Handling
25. Move Shipping and Handling to Shipping Statement
26. Add Merchandise Total, Tax, and Shipping and Handling giving Order Total
27. Move Order Total to Shipping Statement
```

FIGURE 9.6

Structured English for creating the shipping statement for World's Trend.

The final ingredient that makes the decision table worthwhile is the set of rules for each of the actions. Rules are the combinations of the condition alternatives that precipitate an action. For example, Rule 3 says:

IF	N	(the total sale is NOT under $50.00)
		AND
IF	Y	(the customer paid by check and had two forms of ID)
		AND
IF	N	(the customer did not use a credit card)
		THEN
DO	X	(call the supervisor for approval).

The foregoing example features a problem with four sets of rules and four possible actions, but that is only a coincidence. The next example demonstrates that decision tables often become large and involved.

FIGURE 9.7

The standard format used for presenting a decision table.

Conditions and Actions	Rules
Conditions	Condition Alternatives
Actions	Action Entries

FIGURE 9.8

Using a decision table for illustrating a store's policy of customer checkout with four sets of rules and four possible actions.

Conditions and Actions	Rules 1	2	3	4
Under $50	Y	Y	N	N
Pays by check with two forms of ID	Y	N	Y	N
Uses credit card	N	Y	N	Y
Complete the sale after verifying signature.	X			
Complete the sale. No signature needed.		X		
Call supervisor for approval.			X	
Communicate electronically with bank for credit card authorization.				X

Developing Decision Tables

To build decision tables, an analyst needs to determine the maximum size of the table; eliminate any impossible situations, inconsistencies, or redundancies; and simplify the table as much as possible. The following steps provide the analyst with a systematic method for developing decision tables:

1. Determine the number of conditions that may affect the decision. Combine rows that overlap, such as conditions that are mutually exclusive. The number of conditions becomes the number of rows in the top half of the decision table.
2. Determine the number of possible actions that can be taken. That number becomes the number of rows in the lower half of the decision table.
3. Determine the number of condition alternatives for each condition. In the simplest form of a decision table, there would be two alternatives (Y or N) for each condition. In an extended-entry table, there may be many alternatives for each condition. Make sure that all possible values for the condition are included. For example, if a problem statement calculating a customer discount mentions one range of values for an order total from $100 to $1,000 and another range for greater than $1,000, the analyst should realize that the range from 0 up to $100 should also be added as a condition. This is especially true when there are other conditions that may apply to the 0 up to $100 order total.
4. Calculate the maximum number of columns in the decision table by multiplying the number of alternatives for each condition. If there were four conditions and two alternatives (Y or N) for each of the conditions, there would be 16 possibilities, as follows:

   Condition 1: $\times$ 2 alternatives
   Condition 2: $\times$ 2 alternatives
   Condition 3: $\times$ 2 alternatives
   Condition 4: $\times$ 2 alternatives
   16 possibilities

5. Fill in the condition alternatives. Start with the first condition and divide the number of columns by the number of alternatives for that condition. In the foregoing example, there are 16 columns and two alternatives (Y or N), so 16 divided by 2 is 8. Then choose one

CONSULTING OPPORTUNITY 9.3

Saving a Cent on Citron Car Rental

"We feel lucky to be this popular. I think customers feel we have so many options to offer that they ought to rent an auto from us," says Ricardo Limon, who manages several outlets for Citron Car Rental. "Our slogan is, 'You'll never feel squeezed at Citron.' We have five sizes of cars that we list as A through E:

A	Subcompact
B	Compact
C	Midsize
D	Full-size
E	Luxury

"Standard transmission is available only for A, B, and C. Automatic transmission is available for all cars."

"If a customer reserves a subcompact (A) and finds on arriving that we don't have one, that customer gets a free upgrade to the next-sized car, in this case a compact (B). Customers also get a free upgrade from their reserved car size if their company has an account with us. There's a discount for membership in any of the frequent-flyer clubs run by cooperating airlines too. When customers step up to the counter, they tell us what size car they reserved, and then we check to see if we have it in the lot ready to go. They usually bring up any discounts, and we ask them if they want insurance and how long they will use the car. Then we calculate their rate and write out a slip for them to sign right there."

Ricardo has asked you to computerize the billing process for Citron so that customers can get their cars quickly and still be billed correctly. Draw a decision table that represents the conditions, condition alternatives, actions, and action rules you gained from Ricardo's narrative that will guide an automated billing process.

Ricardo wants to expand the ecommerce portion of his business by making it possible to reserve a car online. Draw an updated decision table that shows a 10 percent discount for booking a car over the Web.

of the alternatives, say Y, and write it in the first eight columns. Finish by writing N in the remaining eight columns, as follows:

Condition 1:	Y	Y	Y	Y	Y	Y	Y	Y	N	N	N	N	N	N	N	N

Repeat this step for each condition, using a subset of the table:

Condition 1:	Y	Y	Y	Y	Y	Y	Y	Y	N	N	N	N	N	N	N	N
Condition 2:	Y	Y	Y	Y	N	N	N	N								
Condition 3:	Y	Y	N	N												
Condition 4:	Y	N														

And continue the pattern for each condition:

Condition 1:	Y	Y	Y	Y	Y	Y	Y	Y	N	N	N	N	N	N	N	N
Condition 2:	Y	Y	Y	Y	N	N	N	N	Y	Y	Y	Y	N	N	N	N
Condition 3:	Y	Y	N	N	Y	Y	N	N	Y	Y	N	N	Y	Y	N	N
Condition 4:	Y	N	Y	N	Y	N	Y	N	Y	N	Y	N	Y	N	Y	N

6. Complete the table by inserting an X where rules suggest certain actions.
7. Combine rules where it is apparent that an alternative does not make a difference in the outcome. For example,

Condition 1:	YY
Condition 2:	YN
Action 1:	XX

FIGURE 9.9

Constructing a decision table for deciding which catalog to send to customers who order only from selected catalogs.

Conditions and Actions	Rules 1	2	3	4	5	6	7	8
Customer ordered from Fall catalog.	Y	Y	Y	Y	N	N	N	N
Customer ordered from Christmas catalog.	Y	Y	N	N	Y	Y	N	N
Customer ordered from specialty catalog.	Y	N	Y	N	Y	N	Y	N
Send out this year's Christmas catalog.		X		X		X		X
Send out specialty catalog.			X				X	
Send out both catalogs.	X				X			

can be expressed as:

Condition 1:	Y
Condition 2:	—
Action 1:	X

The dash [—] signifies that Condition 2 can be either Y or N, and the action will still be taken.

8. Check the table for any impossible situations, contradictions, and redundancies. They are discussed in more detail later.
9. Rearrange the conditions and actions (or even rules) if it makes the decision table more understandable.

A DECISION TABLE EXAMPLE Figure 9.9 is shows a decision table developed using the steps previously outlined. In this example, a company is trying to maintain a meaningful mailing list of customers. The objective is to send out only the catalogs from which customers will buy merchandise.

The managers realize that certain loyal customers order from every catalog and that some people on the mailing list never order. These ordering patterns are easy to observe, but deciding which catalogs to send customers who order only from selected catalogs is more difficult. Once these decisions are made, a decision table is constructed for three conditions (C1: customer ordered from Fall catalog; C2: customer ordered from Christmas catalog; and C3: customer ordered from specialty catalog), each having two alternatives (Y or N). Three actions can be taken (A1: send out this year's Christmas catalog; A2: send out the new specialty catalog; and A3: send out both catalogs). The resulting decision table has six rows (three conditions and three actions) and eight columns (two alternatives × two alternatives × two alternatives).

The decision table is now examined to see if it can be reduced. There are no mutually exclusive conditions, so it is not possible to get by with fewer than three condition rows. No rules allow the combination of actions. It is possible, however, to combine some of the rules, as shown in Figure 9.10. For instance, Rules 2, 4, 6, and 8 can be combined because they all have two things in common:

1. They instruct us to send out this year's Christmas catalog.
2. The alternative for Condition 3 is always N.

It doesn't matter what the alternatives are for the first two conditions, so it is possible to insert dashes [—] in place of the Y or N.

The remaining rules—Rules 1, 3, 5, and 7—cannot be reduced to a single rule because two different actions remain. Instead, Rules 1 and 5 can be combined; likewise, Rules 3 and 7 can be combined.

Checking for Completeness and Accuracy

Checking over your decision tables for completeness and accuracy is essential. Four main problems can occur in developing decision tables: incompleteness, impossible situations, contradictions, and redundancy.

Ensuring that all conditions, condition alternatives, actions, and action rules are complete is of utmost importance. Suppose an important condition—if a customer ordered less than

Conditions and Actions	Rules 1	2	3	4	5	6	7	8
Customer ordered from Fall catalog.	Y	Y	Y	Y	N	N	N	N
Customer ordered from Christmas catalog.	Y	Y	N	N	Y	Y	N	N
Customer ordered from specialty catalog.	Y	N	Y	N	Y	N	Y	N
Send out this year's Christmas catalog.		X		X		X		X
Send out specialty catalog.			X				X	
Send out both catalogs.	X				X			

Conditions and Actions	Rules 1'	2'	3'
Customer ordered from Fall catalog.	—	—	—
Customer ordered from Christmas catalog.	Y	—	N
Customer ordered from specialty catalog.	Y	N	Y
Send out this year's Christmas catalog.		X	
Send out specialty catalog.			X
Send out both catalogs.	X		

FIGURE 9.10

Combining rules to simplify the customer-catalog decision table.

$50—had been left out of the catalog store problem discussed earlier. The whole decision table would change because a new condition, new set of alternatives, new action, and one or more new action rules would have to be added. Suppose the rule is: IF the customer did not order more than $50, THEN do not send any catalogs. A new Rule 4 would be added to the decision table, as shown in Figure 9.11.

When building decision tables as outlined in the foregoing steps, it is sometimes possible to set up impossible situations. An example is shown in Figure 9.12. Rule 1 is not feasible because a person cannot earn greater than $50,000 per year and less than $2,000 per month at the same time. The other three rules are valid. The problem went unnoticed because the first condition was measured in years and the second condition in months.

Contradictions occur when rules suggest different actions but satisfy the same conditions. The fault could lie with the way the analyst constructed the table or with the information the analyst received. Contradictions often occur if dashes [—] are incorrectly inserted into the table. Redundancy occurs when identical sets of alternatives require the exact same action. Figure 9.13 illustrates a contradiction and a redundancy. The analyst has to determine what is correct and then resolve the contradiction or redundancy.

Decision tables are an important tool in the analysis of structured decisions. One major advantage of using decision tables over other methods is that tables help the analyst ensure completeness. When using decision tables, it is also easy to check for possible errors, such as

Conditions and Actions	Rules 1'	2'	3'	4'
Customer ordered from Fall catalog.	—	—	—	—
Customer ordered from Christmas catalog.	Y	—	N	—
Customer ordered from specialty catalog.	Y	N	Y	—
Customer ordered $50 or more.	Y	Y	Y	N
Send out this year's Christmas catalog.		X		
Send out specialty catalog.			X	
Send out both catalogs.	X			
Do not send out any catalog.				X

FIGURE 9.11

Adding a rule to the customer-catalog decision table changes the entire table.

FIGURE 9.12

Checking the decision table for impossible situations is important.

Conditions and Actions	Rules 1	2	3	4
Salary > $50,000/year	Y	Y	N	N
Salary < $2,000/month	Y	N	Y	N
Action 1				
Action 2				

This is an impossible situation.

FIGURE 9.13

Checking the decision table for inadvertent contradictions and redundancy is important.

Conditions and Actions	Rules 1	2	3	4	5	6	7
Condition 1	Y	Y	Y	Y	Y	N	N
Condition 2	Y	Y	Y	N	N	Y	N
Condition 3	—	N	—	—	—	N	Y
Action 1	X			X	X		
Action 2			X			X	
Action 3		X					X

Contradiction (Rules 1, 2, 3) Redundancy (Rules 4, 5)

impossible situations, contradictions, and redundancy. Decision table processors, which take the table as input and provide computer program code as output, are also available.

Decision Trees

Decision trees are used when complex branching occurs in a structured decision process. Trees are also useful when it is essential to keep a string of decisions in a particular sequence. Although the decision tree derives its name from natural trees, decision trees are most often drawn on their side, with the root of the tree on the left side of the paper; from there, the tree branches out to the right. This orientation allows the analyst to write on the branches to describe conditions and actions.

Unlike the decision tree used in management science, the analyst's tree does not contain probabilities and outcomes. In systems analysis, trees are used mainly for identifying and organizing conditions and actions in a completely structured decision process.

Drawing Decision Trees

It is useful to distinguish between conditions and actions when drawing decision trees. This distinction is especially relevant when conditions and actions take place over a period of time and their sequence is important. For this purpose, use a square node to indicate an action and a circle to represent a condition. Using notation makes the decision tree more readable, as does numbering the circles and squares sequentially. Think of a circle as signifying IF, whereas the square means THEN.

When decision tables were discussed in an earlier section, a point-of-sale example was used to determine the purchase approval actions for a department store. Conditions included the amount of the sale (under $50) and whether the customer paid by check or credit card. The four

CONSULTING OPPORTUNITY 9.4

A Tree for Free

"I know you've got a plane to catch, but let me try to explain it to you once again, sir," pleads Glen Curtiss, a marketing manager for Premium Airlines. Curtiss has been attempting (unsuccessfully) to explain the airline's new policy for accumulating miles for awards (such as upgrades to first class and free flights) to a member of Premium's "Flying for Prizes" club.

Glen takes another pass at getting the policy off the ground, saying, "You see, sir, the traveler (that's you, Mr. Icarus) will be awarded the miles actually flown. If the actual mileage for the leg was less than 500 miles, the traveler will get 500 miles credit. If the trip was made on a Saturday, the actual mileage will be multiplied by two. If the trip was made on a Tuesday, the multiplication factor is 1.5. If this is the ninth leg traveled during the calendar month, the mileage is doubled no matter what day, and if it is the 17th leg traveled, the mileage is tripled. If the traveler booked the flight on the Web or through a travel service such as Orbitz or Travelocity, 100 miles are added."

"I hope that clears it up for you, Mr. Icarus. Enjoy your flight, and thanks for flying Premium."

Mr. Icarus, whose desire to board the Premium plane has all but melted away during Glen's long explanation, fades into the sea of people wading through the security lanes without so much as a peep in reply.

Develop a decision tree for Premium Airlines' new policy for accumulating award miles so that the policy becomes clearer, is easier to grasp visually, and hence is easier to explain.

actions possible were: complete the sale after verifying the signature; complete the sale with no signature needed; call the supervisor for approval; or communicate electronically with the bank for credit card authorization. Figure 9.14 illustrates how this example can be drawn as a decision tree. In drawing the tree:

1. Identify all conditions and actions and their order and timing (if they are critical).
2. Begin building the tree from left to right, making sure you list all possible alternatives before moving to the right.

This simple tree is symmetrical, and the four actions at the end are unique. A tree does not need to be symmetrical. Most decision trees have conditions that have a different number of branches. Also, identical actions may appear more than once.

A decision tree has three main advantages over a decision table. First, it takes advantage of the sequential structure of decision tree branches so that the order of checking conditions and executing actions is immediately noticeable. Second, conditions and actions of decision trees are found on some branches but not on others; in contrast, with decision tables, conditions and actions are all part of the same table. Conditions and actions that are critical are connected directly to other conditions and actions, whereas conditions that do not matter are absent. In other words, the tree does not have to be symmetrical. Third, compared with decision tables, decision trees are more readily understood by others in an organization. Consequently, they are more appropriate as a communication tool.

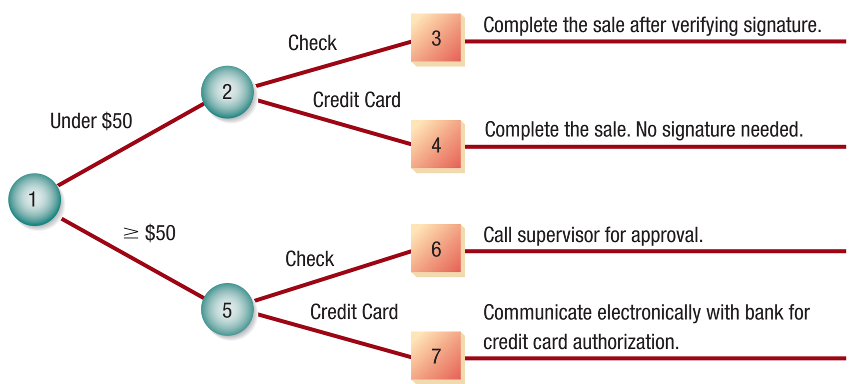

FIGURE 9.14

Drawing a decision tree to show the noncash purchase approval actions for a department store.

Choosing a Structured Decision Analysis Technique

We have examined the three techniques for analysis of structured decisions: structured English, decision tables, and decision trees. Although they need not be used exclusively, it is customary to choose one analysis technique for a decision rather than employing all three. The following guidelines provide you with a way to choose one of the three techniques for a particular case:

1. Use structured English when
   a. There are many repetitious actions,
      OR
   b. Communication to end users is important.
2. Use decision tables when
   a. Complex combinations of conditions, actions, and rules are found,
      OR
   b. You require a method that effectively avoids impossible situations, redundancies, and contradictions.
3. Use decision trees when
   a. The sequence of conditions and actions is critical,
      OR
   b. When not every condition is relevant to every action (the branches are different).

Summary

Once an analyst has worked with users to identify data flows and begun constructing a data dictionary, it is time to turn to process specification and decision analysis. The three methods for decision analysis and describing process logic discussed in this chapter are structured English, decision tables, and decision trees. All three methods help an analyst develop the logic for structured decisions made in the organization. Making structured decisions need not involve human judgment.

Process specifications (or minispecs) are created for primitive processes on a data flow diagram as well as for some higher-level processes that explode to a child diagram. These specifications explain the decision-making logic and formulas that will transform process input data into output. The three goals of process specification are to reduce the ambiguity of the process, to obtain a precise description of what is accomplished, and to validate the system design.

One way to describe structured decisions is to use the method referred to as structured English, in which logic is expressed in sequential structures, decision structures, case structures, or iterations. Structured English uses accepted keywords such as IF, THEN, ELSE, DO, DO WHILE, and DO UNTIL to describe the logic used, and it indents to indicate the hierarchical structure of the decision process.

Decision tables provide another way to examine, describe, and document decisions. Four quadrants (viewed clockwise from the upper-left corner) are used to (1) describe the conditions, (2) identify possible condition (or decision) alternatives (such as Y or N), (3) indicate which actions should be performed, and (4) describe the actions. Decision tables are advantageous because the rules for developing the table itself, as well as the rules for eliminating redundancy, contradictions, and impossible situations, are straightforward and manageable. The use of decision tables promotes completeness and accuracy in analyzing structured decisions.

The third method for decision analysis is to use a decision tree, consisting of nodes (a square for actions and a circle for conditions) and branches. Decision trees are appropriate when actions must be accomplished in a certain sequence. There is no requirement that the tree be symmetrical, so only those conditions and actions that are critical to the decisions at hand are found on a particular branch.

Each of the decision analysis methods has advantages and should be used accordingly. Structured English is useful when many actions are repeated and when communicating with others is important. Decision tables provide a complete analysis of complex situations while limiting the need for change attributable to impossible situations, redundancies, or contradictions. Decision trees are important when proper sequencing of conditions and actions is critical and when each condition is not relevant to each action.

HYPERCASE EXPERIENCE 9

"It's really great that you've been able to spend all of this time with us. One thing's for sure: We can use the help. And clearly, from your conversations with Mr. Evans and others, you must realize we all believe that consultants have a role to play in helping companies change. Well, most of us believe it anyway."

"Sometimes structure is good for a person. Or even a company. As you know, Mr. Evans is keen on any kind of structure. That's why some of the Training people can drive him wild sometimes. They're good at structuring things for their clients, but when it comes to organizing their own work, it's another story. Oh well, let me know if there's any way I can help you."

HYPERCASE Questions

1. Assume that you will create the specifications for an automated project tracking system for the Training employees. One of the system's functions will be to allow project members to update or add names, addresses, email addresses, and phone numbers of new clients. Using structured English, write a procedure for carrying out the process of entering a new client name, address, email address, and phone numbers. (*Hint:* The procedure should ask for a client name, check to see if the name is already in an existing client file, and let the user either validate and update the current client address, email address, and phone number [if necessary] or add a new client's address, email address, and phone number to the client file.)

Keywords and Phrases

action
action rule
condition
condition alternative
decision table
decision tree
minispecs
process specifications
structured decision
structured English

Review Questions

1. List three reasons for producing process specifications.
2. List process categories that do not require specifications.
3. What four elements must be known for a systems analyst to design systems for structured decisions?
4. What are the two building blocks of structured English?
5. List five conventions that should be followed when using structured English.
6. What is the advantage of using structured English to communicate with people in an organization?
7. What is a decision table? Discuss the standard format used to present one.
8. List the steps that provide a systematic method of development for decision tables.
9. List the four main problems that can occur in developing decision tables.
10. How do decision tables play an important role in the analysis of structured decisions to ensure completeness?
11. What are the main uses of decision trees in systems analysis?
12. List the four major steps in building decision trees.
13. Why is it not important for a decision tree to be symmetrical?
14. In which two situations should you use structured English?
15. In which two situations do decision tables work best?
16. In which two situations are decision trees preferable?

Problems

1. Clyde Clerk is reviewing his firm's expense reimbursement policies with the new salesperson, Trav Farr. "Our reimbursement policies depend on the situation. You see, first we determine if it is a local trip. If it is, we only pay mileage of 45 cents a mile. If the trip was a one-day trip, we pay mileage and then check the times of departure and return. To be reimbursed for breakfast, you must leave by 7:00 A.M., lunch by 11:00 A.M., and have dinner by 5:00 P.M. To receive reimbursement

for breakfast, you must return later than 10:00 A.M., lunch later than 2:00 P.M., and have dinner by 7:00 P.M. On a trip lasting more than one day, we allow hotel, taxi, and airfare, as well as meal allowances. The same times apply for meal expenses." Write structured English for Clyde's narrative of the reimbursement policies.

2. Draw a decision tree depicting the reimbursement policy in Problem 1.
3. Draw a decision table for the reimbursement policy in Problem 1.
4. A computer supplies firm called True Disk has set up accounts for countless businesses in Dosville. True Disk sends out invoices monthly and will give discounts if payments are made within 10 days. The discounting policy is as follows: if the amount of the order for computer supplies is greater than $1,000, subtract 4 percent for the order; if the amount is between $500 and $1,000, subtract a 2 percent discount; if the amount is less than $500, do not apply any discount. All orders made online automatically receive an extra 5 percent discount. Any special order (computer furniture, for example) is exempt from all discounting.

   Develop a decision table for True Disk discounting decisions, for which the condition alternatives are limited to Y and N.
5. Develop an extended-entry decision table for the True Disk company discount policy described in Problem 4.
6. Develop a decision tree for the True Disk company discount policy in Problem 4.
7. Write structured English to solve the True Disk company situation in Problem 4.
8. A systems analyst is designing an app that automates loan application processes. However, many factors must be considered before a loan can be processed or verified by the system. The analyst has described the different scenarios as follows:

   > The app first checks whether the loan applicant is an existing customer or a new one. For existing customers, the system first verifies the age of the customer, followed by their source of income, the amount of monthly income, and the loan applied for. It then displays whether the application has been processed successfully.
   >
   > If the age of an applicant is less than 30 and they receive a monthly salary of up to $30,000, their eligibility ranges between a minimum of $200,000 for no less than five years' duration and a maximum amount of $500,000, payable in at least 10 years. Those who earn more than $30,000 a month are eligible for $500,000, payable in at least five years, and a maximum amount of $1,000,000, payable in at least 10 years. For those who earn from businesses, similar slabs apply, but an additional document verification and a registered TaxPayerID is required. Finally, those who earn less than $10,000 a month are not eligible for a loan.
   >
   > All the applicants aged between 30 and 60 years are eligible for a maximum amount of $500,000 for a minimum duration of five years. For those who have retired, the eligible amount is capped at $200,000, payable in at least five years.
   >
   > For new customers, their credit rating is verified first. If found acceptable, they too are eligible for loan applications according to the slabs mentioned above. The application is rejected if their credit rating is unacceptable.

   Draw a decision tree for requirement specification of the application.
9. Write structured English for the loan application process in Problem 8.
10. An e-retailer is famous for making their clearance sales very lucrative through offers bundled together. However, it is sometimes difficult to decide what the offers should be, so the manager has briefly explained how offers currently work at their store as follows:

    > If customers purchase items worth $2,000 and above, they are eligible for a flat 10 percent discount. When they choose to shop from select clearance merchandise and the cart value exceeds $2,000, they get products chosen at the clearance price, an additional 10 percent discount, and a 5 percent cashback.
    >
    > If the customer increases the cart value to $5,000, then the discounts offered are based on the mode of payment chosen:
    >
    > If they pay through any of the listed online modes, they get products listed at the clearance price and an additional 20 percent off on the cart value. However, if they choose cash on delivery, they get an additional 5 percent off.
    >
    > If customers choose both clearance and fresh stocks and build up a cart value exceeding $5,000, they get the clearance stock at clearance prices and the fresh stock at a flat 20 percent off. If they also choose to share this discount and cart details with friends, they get cash coupons worth $5,000.
    >
    > Customers who buy clearance merchandise worth $10,000 or above get a loyalty badge qualifying them for a flat 20 percent discount on the entire stock for a month.

    Which process specification technique would be the most suitable for this scenario? Justify your answer.

Conditions and Actions	Rules															
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Sufficient quantity on hand	Y	Y	Y	Y	Y	Y	Y	Y	N	N	N	N	N	N	N	N
Quantity large enough for discount	Y	Y	Y	Y	N	N	N	N	Y	Y	Y	Y	N	N	N	N
Wholesale customer	Y	Y	N	N	Y	Y	N	N	Y	Y	N	N	Y	Y	N	N
Sales tax exemption filed	Y	N	Y	N	Y	N	Y	N	Y	N	Y	N	Y	N	Y	N
Ship items and prepare invoice	X	X	X	X	X	X	X	X								
Set up backorder									X	X	X	X	X	X	X	X
Deduct discount	X	X														
Add sales tax		X	X	X		X	X	X								

FIGURE 9.EX1

A decision table for a warehouse.

11. Reduce the decision table in Figure 9.EX1 to the minimum number of rules.
12. Azure Isle Resort has a pricing structure for vacationers in one of its three dwelling categories: the hotel, villas, and beach bungalows. The base price is for staying in the hotel. Beach bungalows have a 10 percent surcharge and villas have a 15 percent surcharge. The final price includes a discount of 4 percent for returning customers. Further conditions apply to how close to capacity the resort is and whether the requested date is within one month from the current date. If the resort is 50 percent full and the time is within one month, there is a 12 percent discount. If the resort is 70 percent full and the time is within one month, there is a 6 percent discount. If the resort is 85 percent full and it is within one month, there is a 4 percent discount.

    Develop an optimized decision table for the Azure Isle Resort pricing structure.
13. Create a decision tree for Problem 12.
14. James, the operations manager at a manufacturing firm, is involved in forecasting exercises as part of his job. In one such recent exercise, he found an exponential increase in demand forecasts. To meet the increase in demands for the next two years, he pondered over a few options available to him. One was to upgrade the firm's manufacturing facility with new technology to introduce advancements in design and production. He estimated that one such facility could be set up at a cost of $5 billion, and he would need at least three such facilities. The second option is to expand the existing capacity by installing at least five more production plants at a cost of $2 billion each. The third option is to outsource the production to a third party, which would amount to approximately $25 million. However, James also calculated the difference in revenue estimated for each option, taking into account differences in demand estimates. He categorized demand estimates so that those resembling the forecast are labeled "high" and those resembling past estimates are labeled "moderate." For less than these two levels, he estimated no need for change in the current production cycle. For the first option, if the demand is high, the revenue is $30 billion; if it is moderate, the revenue is $25 billion. For the second option, if the demand is high, the revenue is $7 billion per plant; if moderate, $2 billion. Finally, for the third option, high demand would fetch $30 million in revenue while moderate demand would fetch $20 million.

    Develop an optimized decision table that clearly lists each available option for the manufacturing firm to meet changes in demand.
15. Develop a decision tree for the situation in Problem 14.

Group Projects

1. Each group member (or each subgroup) should choose to become an "expert" and prepare to explain how and when to use one of the following structured decision techniques: structured English, decision tables, or decision trees. Each group member or subgroup should then make a case for the usefulness of its assigned decision analysis technique for studying the types of structured decisions made by Maverick Transport on dispatching particular trucks to particular destinations. As background for the problem, your group should know that Maverick Transport (first seen in Chapter 5) is a 15-year-old national U.S. trucking firm. Maverick is a less-than-a-truckload (LTL) carrier. The people in management work from the philosophy of just in time (JIT), in which they have created a partnership that includes the shipper, the receiver, and the carrier (Maverick Transport) for the purpose

of transporting and delivering the materials required just in time for their use on the production line. Maverick maintains 626 tractors for hauling freight and has 45,000 square feet of warehouse space and 21,000 square feet of office space. Each group should make a presentation of its preferred technique for making structured decisions.

2. After hearing each presentation, the entire group should reach a consensus on which technique is most appropriate for analyzing the dispatching decisions of Maverick Transport and why that technique is best in this instance. Write the group's recommendation for a decision technique in a paragraph.

Selected Bibliography

Anderson, D. R., D. J. Sweeney, T. A. Williams, J. D. Camm, and J. J. Cochran. *An Introduction to Management Science*, 13th ed. Boston, MA: Cengage Learning, 2013.

Good, P. "What Is Structured English?" www.wisegeek.com/what-is-structured-english.htm. Last accessed August 30, 2017.

Kress-Gazit, H., G. E. Fainekos, and G. J. Pappas. "Translating Structured English to Robot Controllers." *Advance Robotics* 22 (2008): 1343–1359.

Wood, H. "An Intro to Logic Trees and Structured Programming." http://www.techopedia.com/2/28552/it-business/enterprise-applications/an-intro-to-logic-trees-and-structured-programming. Last accessed August 30, 2017.